

• Machine Learning Engineer & AI Researcher with 7+ years of experience in reinforcement learning, large language models, and scalable ML systems. Awarded Best MSc Thesis (CAIAC 2024) for advancing model-based reinforcement learning. Previously built AI-driven ranking and search systems serving 40M+ users. Published in ICML, NeurIPS, and CoLLAs, with expertise in deploying ML at scale. Passionate about bridging research and real-world applications, from LLM alignment to decision-making AI.

Experience

- Mila - Quebec AI Institute** — *ML Research Assistant* Montreal, Canada — Dec 2023 – Present
 - Conducting research on AI alignment and interpretability by steering LLMs through activation space manipulation.
 - Exploring scalable deep reinforcement learning architectures and world modeling using large foundation models for model-based RL.
- Mila - Quebec AI Institute** — *MSc Research Student* Montreal, Canada — Jan 2021 – Dec 2023
 - Researched model-based reinforcement learning and various forms of adaptation in continual learning, publishing in top AI conferences.
- CafeBazaar Inc.** — *Data Scientist* Tehran, Iran — Sep. 2017 – Jan. 2021
 - CafeBazaar is Iran’s largest third-party Android marketplace, **serving 40M+ users**.
 - Personalization Team:
 - Developed and deployed an app recommender system, improving **install rates from 1.6% to 2.5%**.
 - Designed a diversification strategy to broaden recommended app categories, enhancing user engagement.
 - Search-ads Team:
 - As a core founding member, led data science, mechanism design, and optimization for the search advertising system. ([Technical Report](#))
 - Helped establish Search-Ads as a top 3 daily revenue driver for the company.
 - Search Team:
 - Improved the app ranking system and built NLP-based services like spell-checking and query suggestion.
 - Reduced the **search bounce rate from 41% to 35%**, increasing search satisfaction.
- Chair for CAMPAR** — *Research Assistant* Munich, Germany — Jul. 2019 – Oct. 2019
 - Worked on 3D camera re-localization in dynamic environments. Developed a valid pose generator using rendered scenes for 3RScan data.

Education

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| • University of Montreal - Mila
M.Sc. in Computer Science (Artificial Intelligence) | • Sharif University of Technology
B.Sc. in Computer Science |
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Selected Publications * DENOTES EQUAL CONTRIBUTION.

- Steering Large Language Model Activations in Sparse Spaces**
Under Review ([paper](#))
Rahimi-Kalahroudi, A.*, Bayat, R.*, Pezeshki, M., Chandar, S., Vincent, P.
- Replay Buffer with Local Forgetting for Adapting to Local Environment Changes in Deep MBRL**
Conference on Lifelong Learning Agents, **CoLLAs 2023 (Oral)** ([paper](#))
and **NeurIPS 2022** - Deep Reinforcement Learning Workshop
Rahimi-Kalahroudi, A., Rajendran, J., Momennejad, I., Van Seijen, H., Chandar, S.
- Towards Evaluating Adaptivity of Model-Based Reinforcement Learning Methods**
International Conference on Machine Learning, **ICML 2022** ([paper](#), [code](#))
Rahimi-Kalahroudi, A.*, Wan, Y.*, Rajendran, J., Momennejad, I., Chandar, S., Van Seijen, H.

Honors and Awards

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| • Best M.Sc. Thesis - Canadian Artificial Intelligence Association (CAIAC) | Guelph, Canada — 2024 |
| • Exceptional Grade for the M.Sc. Thesis (Top 1%) - University of Montreal | Montreal, Canada — 2023 |
| • Bourse D’excellence DIRO - University of Montreal | Montreal, Canada — 2022 |
| • M.Sc. Tuition Exemption Scholarships (Bourse C) - University of Montreal | Montreal, Canada — 2021 |
| • International Undergraduate Excellence Award - Technical University of Munich | Munich, Germany — 2019 |
| • Silver Medal - 24 th Iranian National Olympiad in Informatics (INOI) | Tehran, Iran — 2014 |

Skills

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| • Programming: Python, Scala, and C++. | • Data Technologies: Apache Spark, Hadoop, Airflow, and Kafka. |
| • ML Tools: PyTorch, Tensorflow, NumPy, and JAX. | • Misc: Docker, Apptainer, Bash, Slurm, and L ^A T _E X |